INTRODUCTION TO JAVASCRIPT

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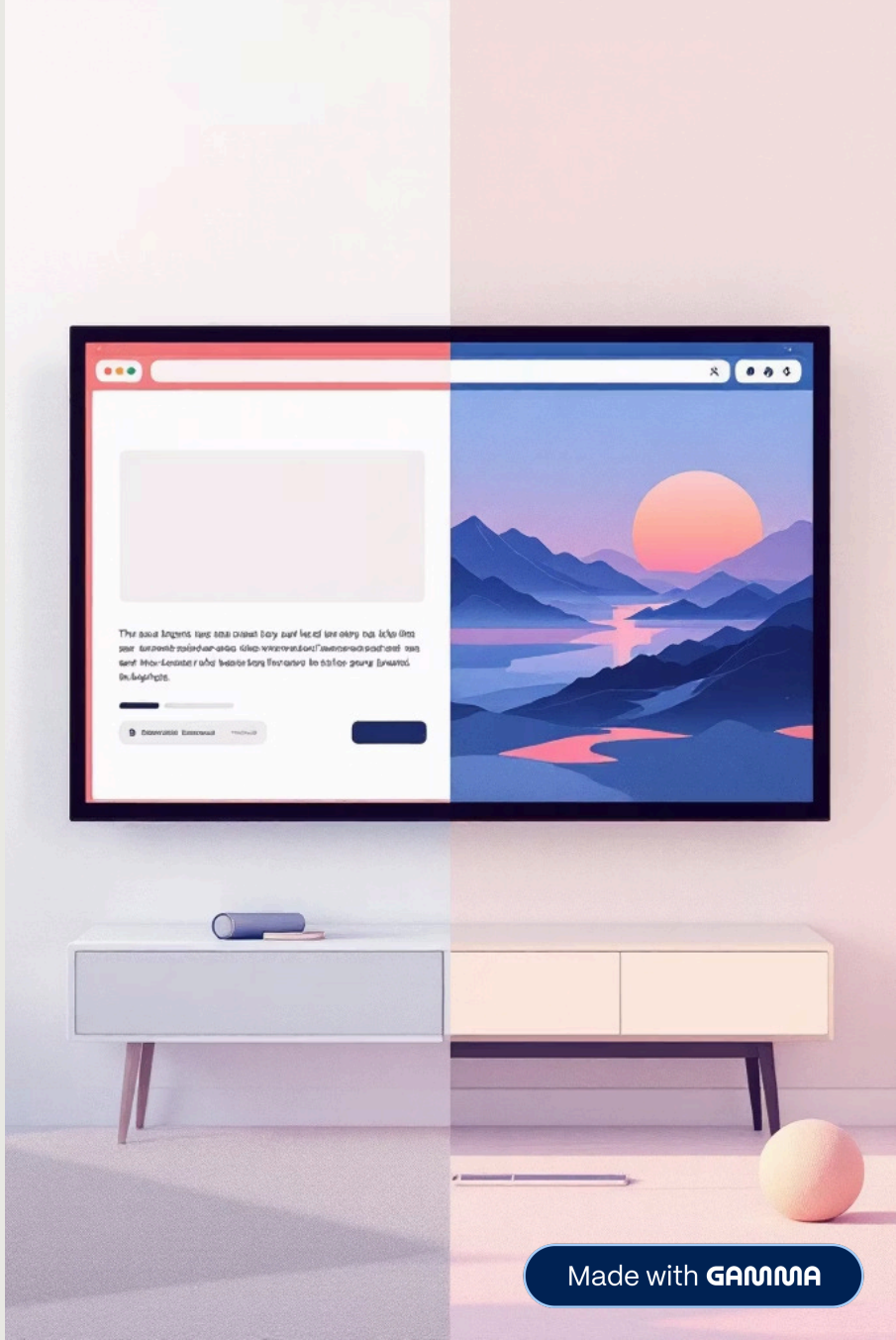
WHAT IS JAVASCRIPT?

HIGH-LEVEL & INTERPRETED

Readable syntax, runs without compiling

MAKES WEBSITES INTERACTIVE

Responds to clicks, typed input, timers



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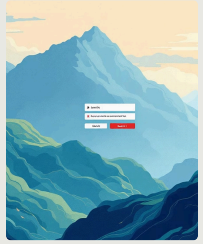
WHY USE JAVASCRIPT IN THE FRONT-END?

HTML + CSS (STATIC)

- Show content and style
- No behavior on its own

JAVASCRIPT (INTERACTIVE)

- Handles clicks & input
- Updates page without reload



INTERACTIVE FORMS

Validate input before submit

BUTTONS

Trigger actions instantly

DYNAMIC COUNTERS

Live updates on the page

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WHERE DOES JAVASCRIPT RUN?

IN THE BROWSER

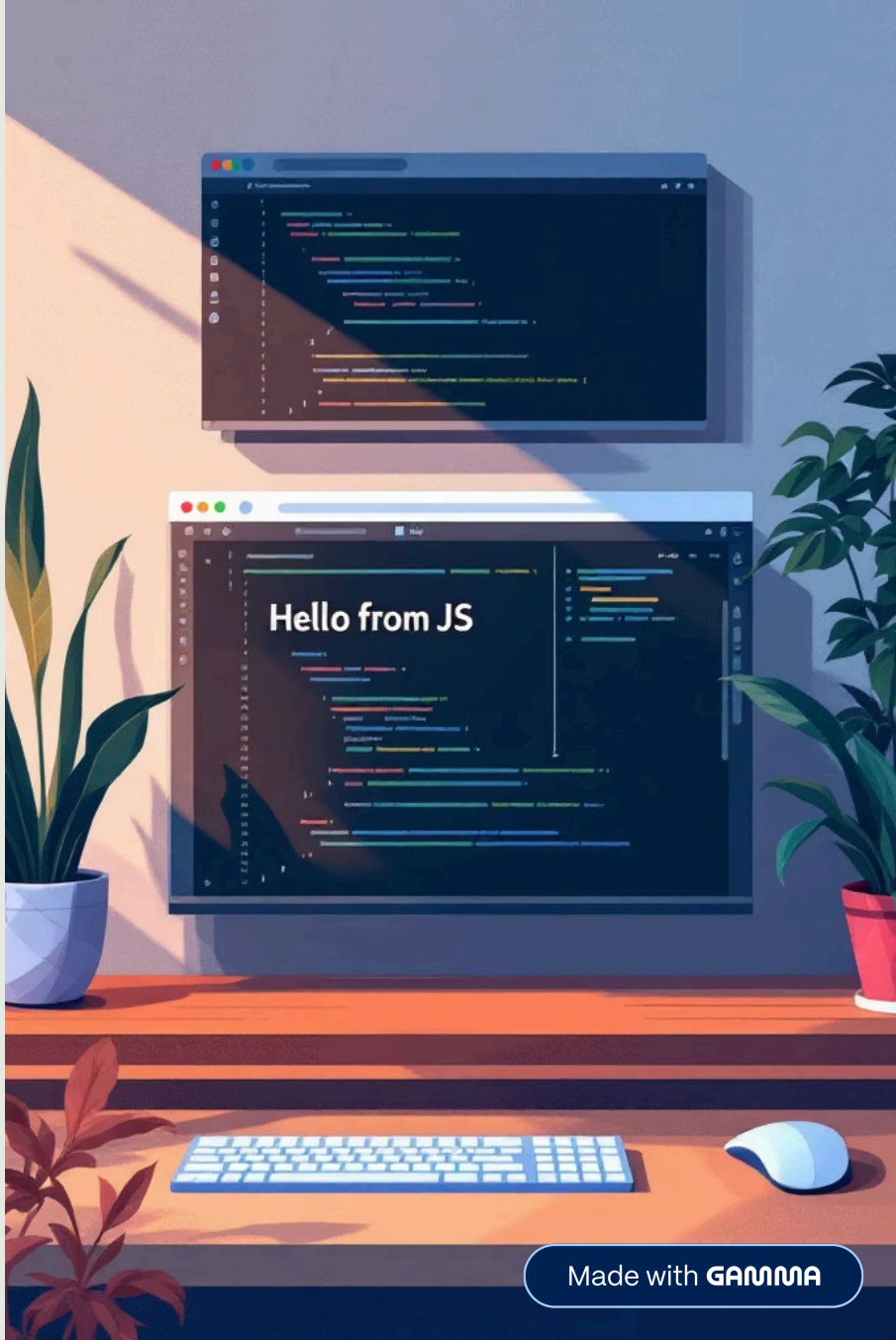
Every browser has a JS engine (e.g., V8 in Chrome)

CONSOLE & SCRIPT TAGS

Type code in Console or add <script> in HTML

ENGINES DO THE WORK

They read and run your JS code



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NODE.JS – JAVASCRIPT OUTSIDE THE BROWSER

WHAT IS NODE.JS?

open-source, cross-platform JavaScript **runtime environment**

- Not a framework
- Built on the V8 engine

BACKEND

Build APIs and servers

TOOLS

Build tools (bundlers, dev servers)

EVERYWHERE

Same language front-to-back

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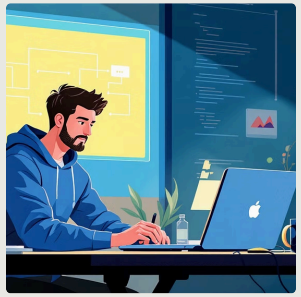
JS AND FRAMEWORKS / LIBRARY

BROWSERS UNDERSTAND:

- HTML
- CSS
- JavaScript

REACT



VUE

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KEY TAKEAWAYS

WHAT JS IS

High-level, interpreted language for web interactivity

WHAT IT DOES

Makes pages respond to users (clicks, typing, timers)

WHERE IT RUNS

Browser engines (V8) or Node.js on servers

FRAMEWORKS

Tools that build on JS

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